

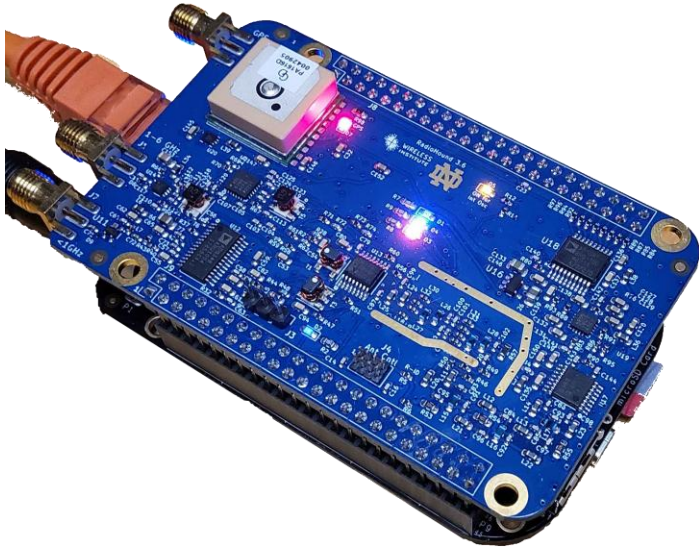
Welcome Fall 2026 STRAND MentEEs

We're excited for a great semester!

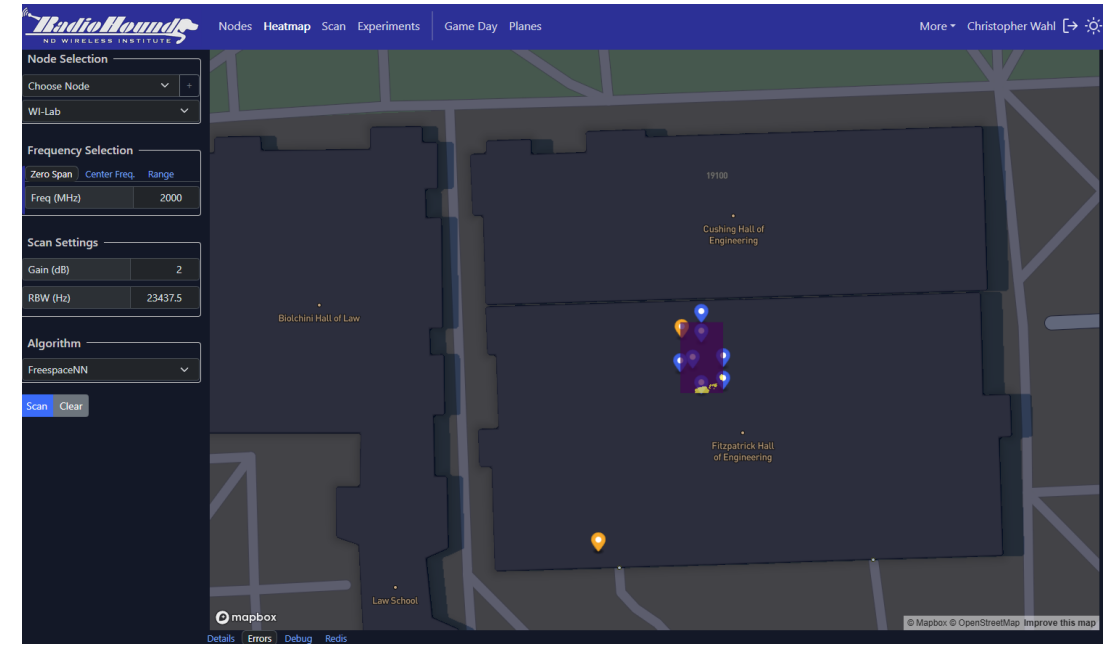
1. RadioHound
2. Going Into Fall 2026
3. Map
4. Things I'm Required to Say
5. Some Final Thoughts

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RadioHound, est. 2015



Sensors: Spectrum sensing requires sensors



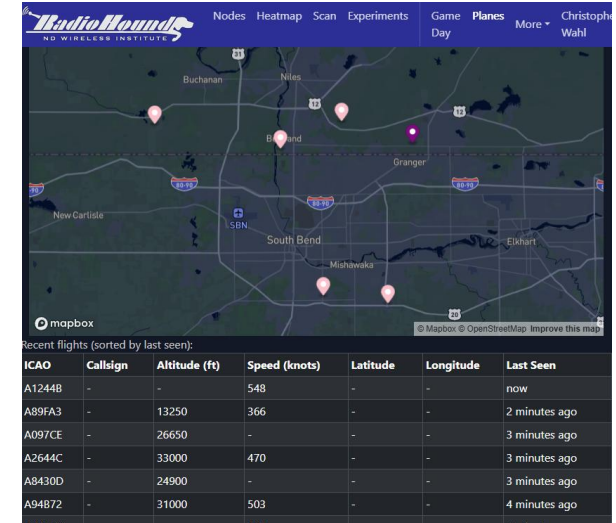
Website/Icarus: Command, control, and visualization



Database: Storing measurements for future research

RadioHound thus Far has Achieved Many Things...

1. Graduate student run project
2. Platform for Spectrum [and other] Sensors
3. Live spectrum sensor data visualization
4. Distributed radio map estimation
5. Real-World spectrum monitoring (ADSB, ND Stadium)



...And Now We're Moving Toward Bigger & Better

This is where you all come in!

1. Time-synchronized distributed sensors
2. Wideband spectrum sensing
3. Real-time data processing
4. `radiohound.run()`

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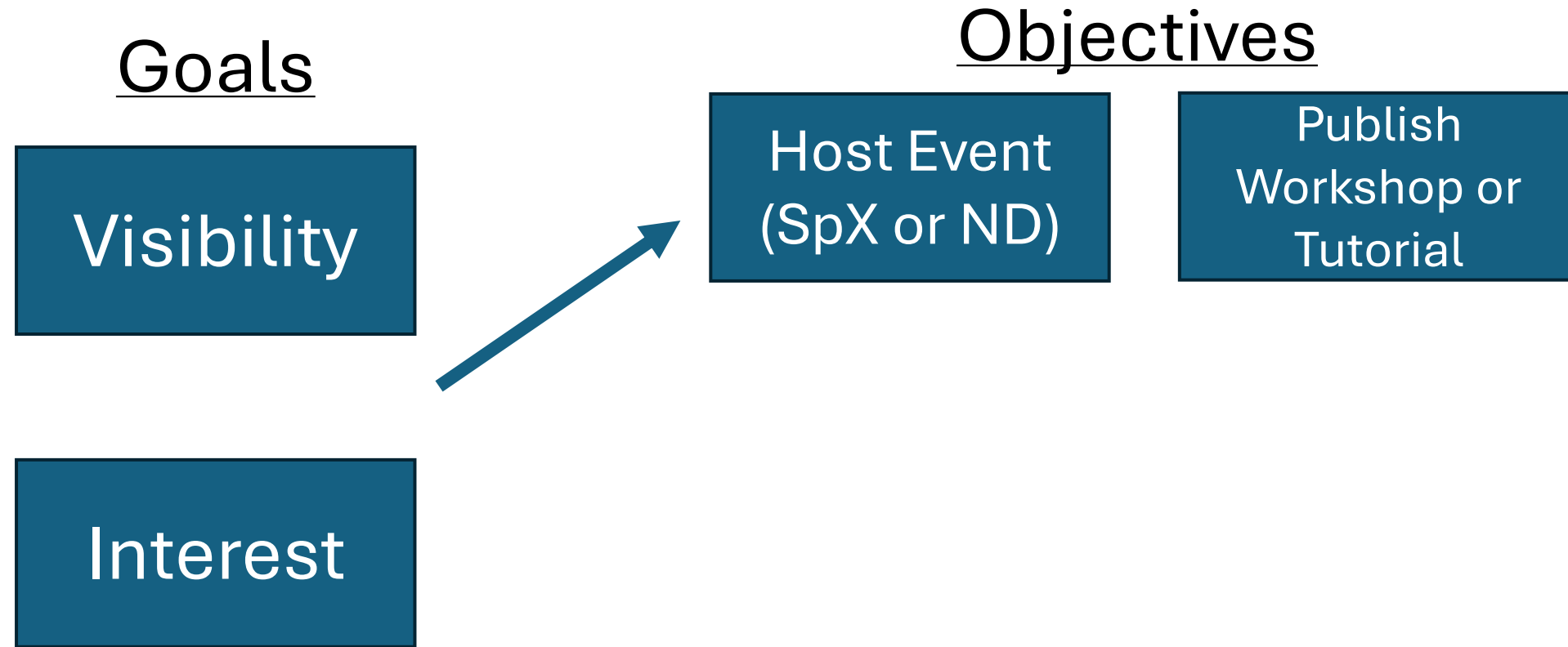
Goals, Objectives, and Strategy for Fall 2026

Goals

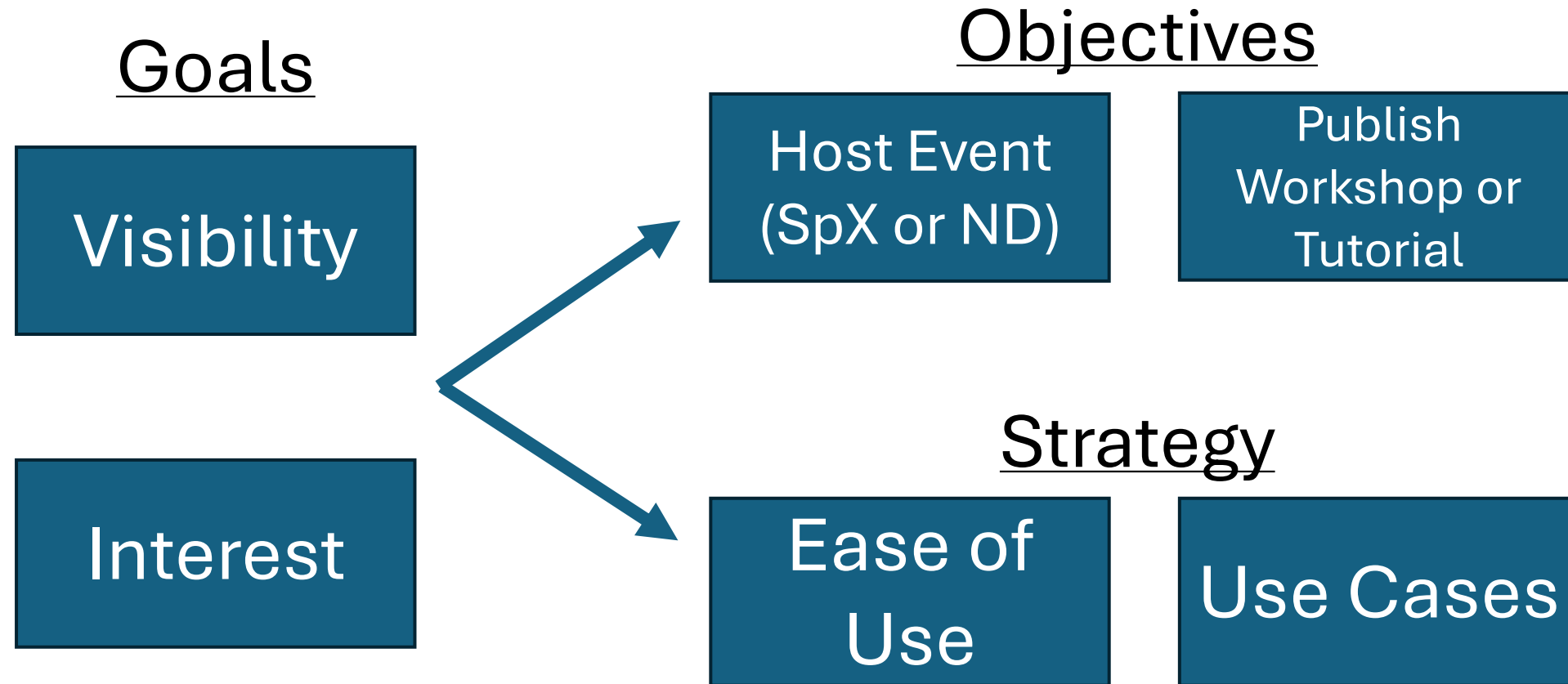
Visibility

Interest

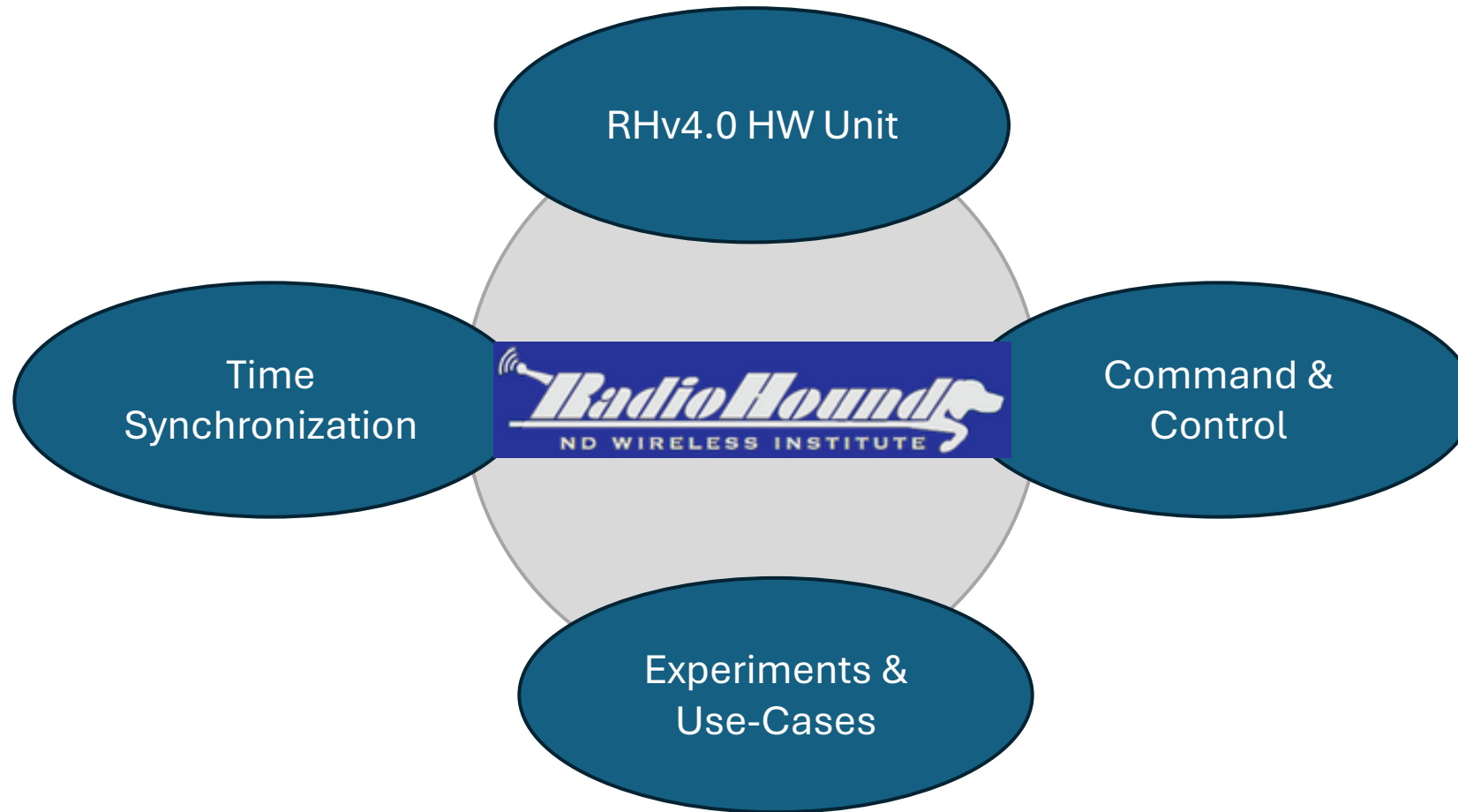
Goals, Objectives, and Strategy for Fall 2026



Goals, Objectives, and Strategy for Fall 2026



Action Items for Fall 2026



Action Items for Fall 2026

RHv4.0 HW Unit

Fall 2026:

- Design radio board(s)
- Automate calibration

Looking Ahead into Spring 2027:

- Design with smaller form factor
- Find an IP66 (or better) enclosure
- Find/Design peripherals
 - Battery
 - Solar
 - Info Screen

STRAND Allocations:

Radio Board(s): Kevin w/ Chris

- RHv4.0 Board in-hand

Calibration: Alonso w/ Chris

- Automated calibration procedure (w.r.t. the antenna port)

FPGA DAQ: Andres, Nico, Michael w/ Dom & Zhiyu

- Raw capture, FFT, and Periodogram modes are switchable
- Ring Buffers
- CPU-side Python interface

FPGA DAQ 2: Luke, Heather w/ Dom & Zhiyu

- Raw capture, FFT, and Periodogram modes are switchable
- Ring Buffers
- CPU-side Python interface

Action Items for Fall 2026

Time Synchronization

Fall 2026:

- Align time samples with GPS PPS
- Align different node CPU times with PTP

Looking Ahead into Spring 2027:

- Perform “Coordinated” jobs/captures between nodes

STRAND Allocations:

GPS PPS: Jack w/ Dom or Zhiyu

- Time stamped samples from the radio through the FPGA

Ethernet PTP: Cole w/ Nazim

- [Best effort] time stamped samples from radio through PTP

Action Items for Fall 2026

Command & Control

Fall 2026:

- Update Icarus for the new hardware
- Design LoRa backhaul

Looking Ahead into Spring 2027:

- Continue updating Icarus?
- Integrate LoRa into Icarus
- Investigate what LoRa can support

STRAND Allocations:

Cellular: Maddy w/ Yankai

- Decide on a cellular module (Quectel)
- Build Quectel module in RHv4.0 Unit
- Write interface code for the cellular module

LoRa: Ryan w/ Yankai

- Decide on a LoRa module
- Build LoRa module into the RHv4.0 Unit
- Write interface code for the LoRa module

Action Items for Fall 2026

Experiments & Use Cases

Fall 2026:

- Run the DeBart Quad RME data experiment
 - Need to make a slide for this
- Build and place dedicated RH node spots in the lab for demos
- Run heterogeneous occupancy map
- Investigate use cases for RHv4.0

Looking Ahead into Spring 2027:

- Build “radiohound” Python module
- Design Software-Defined Experiments

STRAND Allocations:

DeBart Quad Experiment: Jordan w/ Chris

- Conduct a propagation experiment in the DeBart Quad for testing/validating occupancy map estimation

Dedicated RH Lab Spots: Patrick w/ Randy

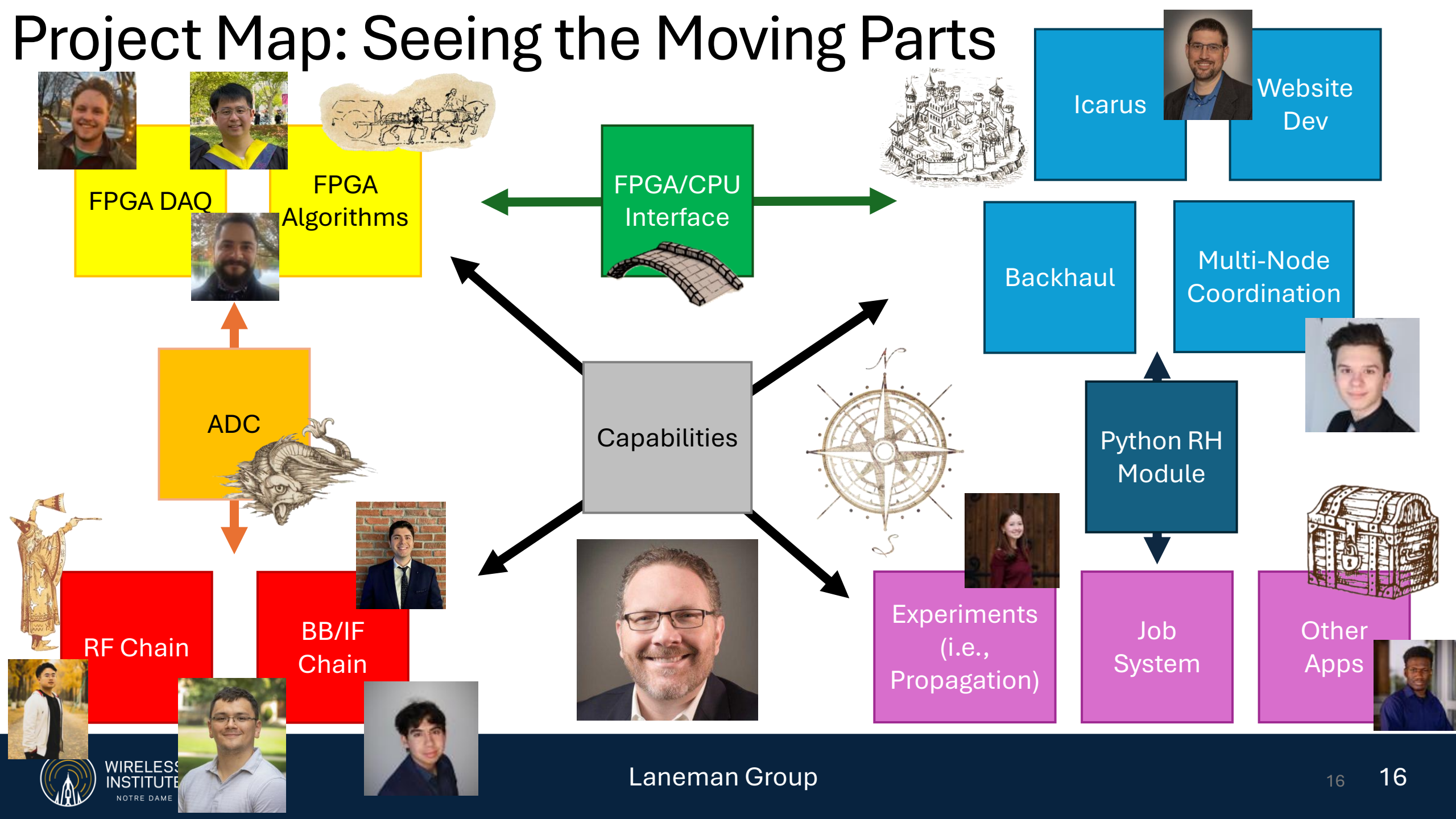
- Build and wire RH Node spots in the lab (Randy’s blue plaque)

Investigate Use Cases: Ava, Ellie, Elias w/ Timi and Yankai

- Ellie works with the City of South Bend
- Ava has worked with satellites before
- Elias will filter interesting things

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Project Map: Seeing the Moving Parts



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Expectations for Fall 2026

1. Project Ownership

- STRAND mentees are expected to “own” their projects
- *Taking accountability vs. Completing tasks* (REU Alonso)
- Prof. Laneman and the grad students are here to help!
- When you succeed, we succeed

2. Time Commitment

- Each credit hour is a minimum of 4 hours of work a week
- **Highly** encouraged to do at least half of your time **in the lab**
- *You get out what you put in*

3. Deliverables

- Milestones
- Documentation and Final Presentation
- Other STRAND Deliverables (required by the STRAND program)

4. Keys

- Keys to the lab *can* be given as needed
- Reserved for mentEEs requiring lab work
- 249/250 is typically open 9-6
- Can always request time outside normal hours

5. Provide Weekly Updates

- Keep your mentor updated at least weekly
- Not a formal deliverable

6. Lab Safety Training

- I need to send out safety training for everyone
- I need your NDID starting with 9
- I know this training is 95% about chemicals
- I will spend more time on the backend setting up your training than you spend taking it
- Do it anyways 😊

7. Google Calendar

- This lab operates on Google Calendar
- If you need something, schedule it

Important Dates for Fall 2026

Problem Description and Proposed Solution: Thursday September 10th, 2026

- Describe the problem you're solving
- Propose a solution
- Explain why this is your solution
 - Is it easy to use? Does it solve the described problem?

Midterm Update: Thursday October 15th, 2026

- Where are you now? Has anything about the proposed solution changed?
- What are your steps moving forward?
- What is your current timeline?

Final Presentation & Documentation: Wednesday December 9th, 2026

- What did you solve?
- Were you successful in solving your problem?
 - If yes, how did you do?
 - If no, how far did you get? Does it look promising?
- What are the advantages of your system? What are the drawbacks?
- What are your recommended next steps?
- Where is the documentation?

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Things to Keep in Mind when Starting this Semester

1. Embrace the Boring Ideas

- Sometimes the obvious solution is the best solution

2. Get a Sponsor

- i.e., You'll [logistically] have an easier time if you get Prof. Laneman or a grad student on-board and excited about what you're doing

3. Set your Scope

- Understand your strengths (and weaknesses)
- Be clear where your planned work starts and ends (w.r.t. RadioHound)
- Define what assumptions you are making

4. Make Sure it's Achievable

- You all have 3-4 months working part-time
- You all are still engineers-in-training

5. Leverage the Graduate Students and Prof. Laneman

- We are here to help and teach you all
- We are all on the same team!
- “You cannot hope to build a better world without improving the individuals” – Marie Curie

6. There are **No** Stupid Questions

- Chris gives inverted clock story example



(Someone smarter than me)

Points 1-4 are taken from “The Staff Engineer’s Path” by Tanya Reilly – Highly recommend

Why Choose the RadioHound STRAND?

“The STRAND of men and women without fear”
– Napoleon (probably)



“La Batterie des Hommes sans Peur” in 1793 during the Siege of Toulon

Why Choose the RadioHound STRAND?



“The STRAND of men and women without fear” – Napoleon (probably)

You get:

1. To contribute to an engineer project
2. Experience working on a long-term technical project
3. Freedom and support to explore [wireless] interests
4. Opportunity to learn engineering soft skills (e.g., project ownership)
5. To network with academics and industry professionals
6. Meet Chris Wahl

“La Batterie des Hommes sans Peur” in 1793 during the Siege of Toulon

Thank You!
Any questions?

This presentation was made without the use of AI.